

Bedfordview Primary School
TERM 4 TEST

Name: _____
Subject: **Natural Sciences & Technology**
Examiner: **Miss Z Kajee & Mrs N Suliman**
Moderator: **Mr J Nash**

Grade 6 : ____

Date: **November 2023**
Time Allocation: **1 ½ hours**

INSTRUCTIONS

1. An extra 10 minutes will be provided for reading.
2. Write neatly and legibly.
3. Answer in the spaces provided.
4. This paper consists of 12 pages.

Parent's signature: _____

Comments: _____

	Section A				Section B		Teacher comment:	
	/30				/30		_____	
Question	1	2	3	4	5	6	TOTAL	%
Allocated mark	4	4	19	3	20	10	60	
Learner's mark								
Moderated mark								
1 0 – 29%	2 30 – 39%	3 40 – 49%	4 50 – 59%	5 60 – 69%		6 70 – 79%	7 80 – 100%	

SECTION A – ENERGY AND CHANGE

[30 Marks]

Question 1: Definitions

(4)

Match the term in **Column A** with its description in **Column B**. Write the correct letter next to the question number in the space provided.

Column A	Column B
1.1 Hydro-electric power	A. The network of electricity cables that transport electricity from the power station to our homes.
1.2 Non-renewable energy	B. Energy resources that can be produced continuously such as wind and hydroelectric energy.
1.3 Renewable energy	C. Energy sources that cannot be replaced once used such as coal; oil; gas and nuclear fuels.
1.4 National grid	D. Electricity produced using the energy from fast-flowing water.

1.1 _____ 1.2 _____ 1.3 _____ 1.4 _____

Question 2: True or False

(4)

State whether the following statements are **TRUE** or **FALSE**. If false , correct the statement to make it true.

2.1 Materials that resist the flow of electricity are called conductors.

2.2. Fossil fuels are natural fuels which forms from dead animals and plants over millions of years.

2.3. A battery has two terminals; a positive and a negative terminal.

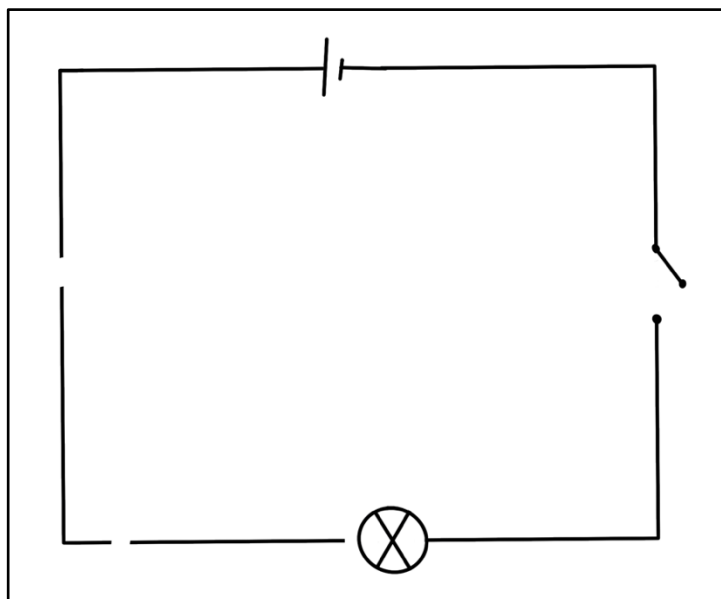
Question 3: Electric Circuits and Systems

(19)

3.1 Study the electric circuit diagram below and answer the questions that follow.

3.1.1 Look at the circuit diagram below. **The bulb does not light up for 4 reasons. Draw a circle around the parts of the circuit that prevents the light bulb from lighting up.**

(4)



3.1.2 Redraw the above circuit diagram **if the light bulb had to shine**. Ensure your diagram has: **A battery of 2 cells, wire, a light bulb, and a switch. You should indicate the positive and negative terminal on the battery and show the flow of current.** (5)

How I Spend my Time During a Test



You can do it! Keep doing your best!



3.2. Read and study the following article and answer the questions that follow.

SOUTH AFRICA

Fearing for their lives, Eskom workers abandon plan to cut illegal connections

15 June 2023 - 18:24

Phathu Luvhengo

Eskom technicians abandoned part of their operation to remove illegal connections in Marlboro industrial area on Thursday for safety reasons. People living illegally in an abandoned factory had allegedly connected electricity illegally and threatened to burn down nearby factories if Eskom cut the cables. The utility's technicians initially disconnected residents in Stjwetla informal settlement in Alexandra, Johannesburg, who had built their homes under Eskom pylons. They had illegally connected cables to a substation in Marlboro Gardens.

One of the residents, Charlotte Motseo from Limpopo, complained that the settlement was not safe without electricity. She insisted Eskom should allow them to connect. She said if she had a job she would have probably stayed in a better place and there would not be a need for residents to illegally connect electricity. She said she would boycott the next year's elections. "How will we vote if they're doing this to us? They could just leave us or electrify our area so that we will be safe. Now they have disconnected us and there is a lot of crime here," she said.

The utility disconnected the area where residents had built shelters under the Eskom 88kV power lines. Technicians disconnected the exposed live wires from the informal settlement.

Just a few metres away from the settlement is the industrial area, where one of the businessmen who had been operating for more than seven years claimed residents had allegedly threatened to burn down his factory. He said people living in one of the abandoned factories connected a cable to a substation that connects to his factory, which manufactures water bottles. "If that cable comes down now, tomorrow morning you will find this factory burnt to the ground," he said. He pleaded with Eskom technicians not to disconnect the cables. He said residents told him several times that if they don't have electricity the entire neighbourhood should be disconnected as well.

For fear for their safety and citing limited law enforcement agencies and a few private security guards who had accompanied them, Eskom technicians eventually abandoned their operation. Eskom senior manager for maintenance and operations in Gauteng, Mashangu Shivambu, said the utility suffered an annual loss of about R7bn in the province due to non-technical issues usually caused by illegal connections.

He said when Eskom disconnected the wires, residents usually reconnected them. "Unfortunately, it is a repeated cycle. We remove it today, and the next day they reinstall — but we can't just fold our arms. "Whatever happens here affects everyone. These illegal networks result in customers being constantly off because of the faults that are generated from these networks," he said. He said it was not only dangerous but also illegal for residents to build homes under Eskom's pylons or high-voltage power lines.

"If you do get a fault on a pylon network, that fault is going to go through those pylons directly to the ground and it becomes very dangerous. People can get killed even if they are not directly connected to the pylons," he said.

TimesLIVE <https://www.timeslive.co.za/news/south-africa/2023-06-15-fearing-for-their-lives-eskom-workers-abandon-plan-to-cut-illegal-connections/>



3.2.1 What was the **reason** for the **operation** performed by the **Eskom workers**? (1)

3.2.2 **Discuss** any **2 dangers** which may result because of the actions of the residents. (2x ½ =1)

3.2.3 **Explain** why people need to **pay** for electricity. (1)

3.2.4 **Why** did the Eskom technicians **abandon** their operation? (1)

3.3. Cost of Electricity:

The grade 6 learners at Bedfordview Primary School investigated which electrical appliances use more electricity to run than others. They looked at the back or bottom of each appliance and recorded the information in the table as shown below.

Study the table and answer the questions that follow.

<u>Appliance Number:</u>	<u>Name of Appliance:</u>	<u>Power Rating in Watt (W):</u>
1.	LED light bulb	20W
2.	Fridge	120W
3.	Heater	100W
4.	Incandescent lightbulb	40W

3.3.1 **Draw a bar graph** showing the power rating in Watts (W) for each appliance to help the grade 6 learners identify which appliance uses more electricity. (6)

Remember to **write all your numbers in blue pen**. Your graph is to be **drawn in pencil**.

Remember to label both the x and y – axis. You must give your graph a suitable heading.

Your graph will be marked accordingly:

- **Heading (1)**
- **Y axis heading (1)**
- **Axis scale (2 x ½ = 1)**
- **X axis heading (1)**
- **Bars (4 x ½ = 2)**



Question 4: Renewable and Non-renewable Energy Sources**(3)**

4.1 The table below shows how coal is formed over millions of years. The steps have been mixed up. Arrange the steps in the correct order by writing only the missing numbers on the right.

(3)

Incorrect order	Correct order
Layers of coal are formed.	4
When the plants died, they were covered under layers of sediments over millions of years, after sinking to the bottom of swamps.	
Millions of years ago there were large forests in swampy areas.	
Pressure and heat compacted the sediment layers.	

TOTAL SECTION A: 30**Turn over for Section B**

SECTION B: PLANET EARTH AND BEYOND**[30 Marks]****Question 5: Planets and our Moon****(20)**

5.1. Using the clues, unscramble the following terminology.

5.1.1. A system consisting of many stars, together with dust and gas.

lxaagy: _____ (1)

5.1.2. The shape in which the planets revolve around the Sun.

pteliillca: _____ (1)

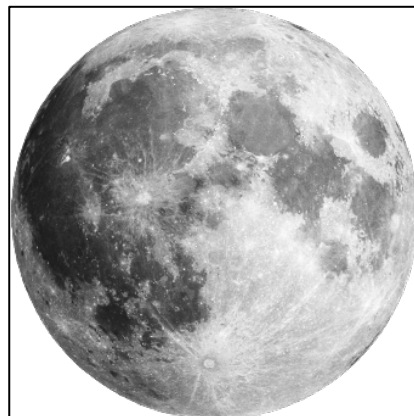
5.1.3. A human-made or natural object that revolves around a planet.

teasiltle: _____ (1)

5.2. Complete the following table based on the amount of time it takes for the planets to revolve around the Sun.

<u>Planet:</u>	<u>Number of Days or Years:</u>
Jupiter	5.2.1 _____ (1)
Neptune	5.2.2 _____ (1)
5.2.3 _____ (1)	365 $\frac{1}{4}$ days

5.3. Look at the picture of the moon below and answer the questions that follow.



5.3.1. **Why** is the moon **cold**?

(2)

(2)

(3)

Using the PEEL method, write a paragraph **explaining the difference between a rotation of the Earth, the revolution of the Earth and how it affects our lives on Earth.**

- What is an axis of the Earth? (1)
- What is a rotation of the Earth? (1)
- How does the rotation of the Earth affect our lives on Earth? (1)
- Explain how we are able to experience daytime and night-time. (2)
- What is the revolution of the Earth around the Sun? (1)
- How does the revolution of the Earth around the Sun affect our lives on Earth? (1)



(10)

	Description/Function of the telescope	Example
Optical Telescope	_____ _____ _____ _____ _____ _____ _____ (1)	_____ _____ (1)
Radio Telescope	_____ _____ _____ _____ _____ _____ _____ (1)	_____ _____ (1)
What is a telescope?		
_____ _____ _____ _____ (1)		



Answer the following questions to build your project on investigating the surface of the moon.

6.2.1. What is a moon rover?

6.2.2. Identify 3 design specifications of your moon rover.

6.2.3. Define the term "astronomer".

TOTAL SECTION B: 30

TOTAL: 60

----- **END OF TEST** -----

